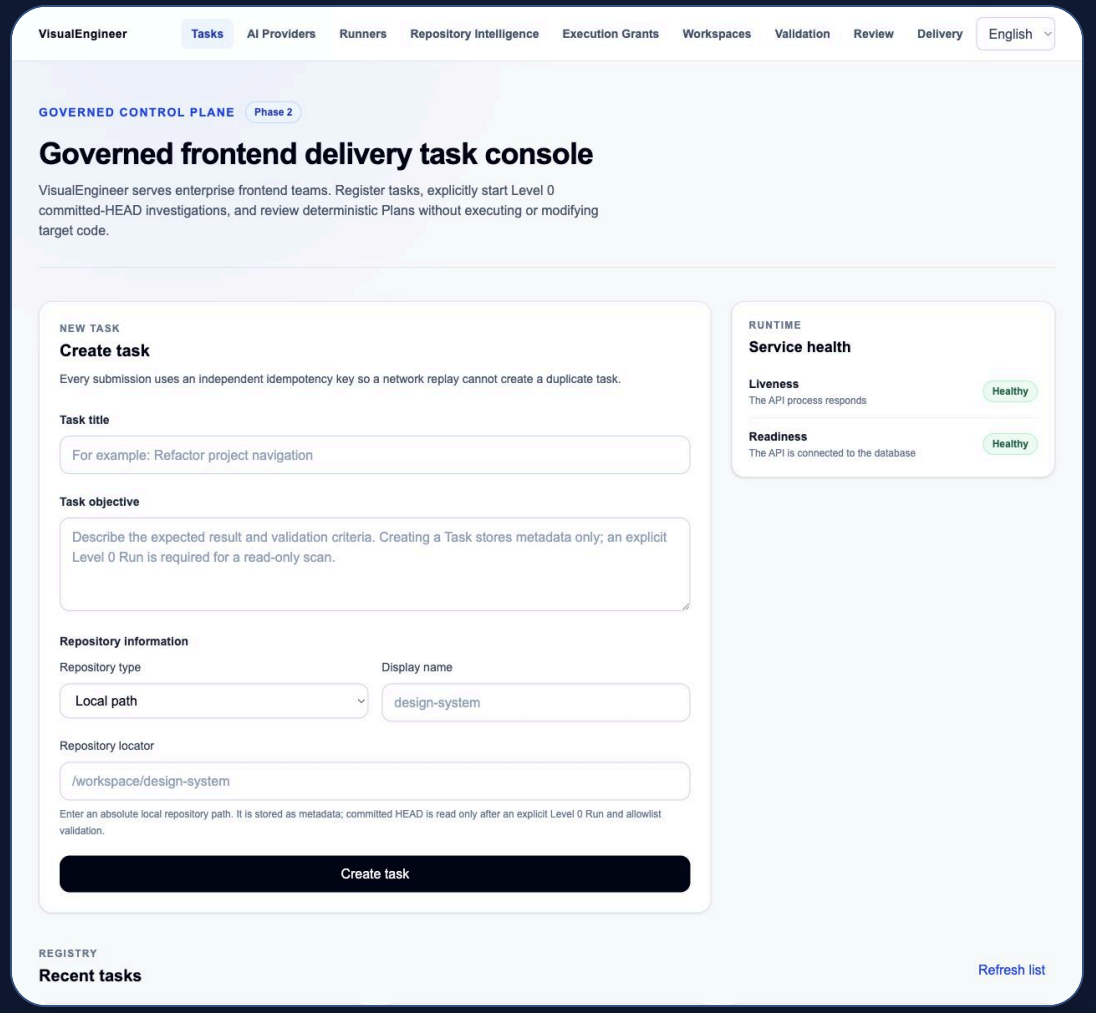
INTERVIEW PORTFOLIO

VISUALENGINEER

Governed AI design-engineering and delivery.

A local, single-user system that keeps AI planning, authorization, execution and evidence as distinct, reviewable decisions.

Engineering case study
github.com/dankk0220abc-prog/visual-engineer



THE PROBLEM

AI can propose implementation work. It should not automatically gain authority to change a repository.

A useful plan is **not** permission to mutate a workspace, run a command, approve a review or ship a delivery.

01 / SEPARATE

Planning from authority

Provider output is typed data. It enters a controlled lifecycle rather than becoming a tool call.

02 / BOUND

Execution to a grant

A distinct scoped grant is required before eligible Runner work in a fixture workspace.

03 / PRESERVE

Uncertainty honestly

Ambiguous external outcomes remain `outcome_unknown`; they are not auto-retried.

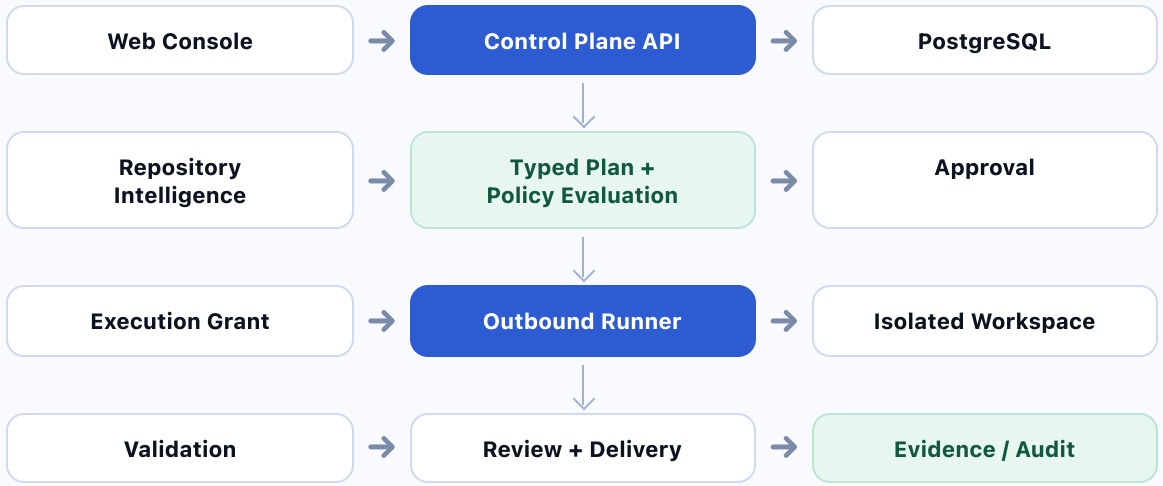
04 / PROJECT

Evidence safely

Reviewable facts are projected without keys, raw traffic, full upstream IDs or billing material.

SYSTEM ARCHITECTURE

A control plane around a bounded AI proposal.



Critical boundary

The Provider Layer may return a typed planning artifact through the Control Plane. It receives no direct Runner or workspace authority.

Responsibilities at a glance

W **Web Console**

Browser-first presentation, workflow and evidence surface.

A **Control Plane**

Lifecycle, policy, authority transitions and safe projection.

R **Local Runner**

Outbound consumer of closed typed jobs and grants.

E **Evidence**

Audit facts suitable for review, without a secret archive.

GOVERNED WORKFLOW

Plan Approval does not equal an Execution Grant.

Provider proposal

A typed Plan is created through the governed path and evaluated by system policy.

Plan generated

Authority boundary

A successful proposal does not decide approval, issue a grant, grant Runner authority, create a review or authorize delivery.

Human decision required

Separate scope required

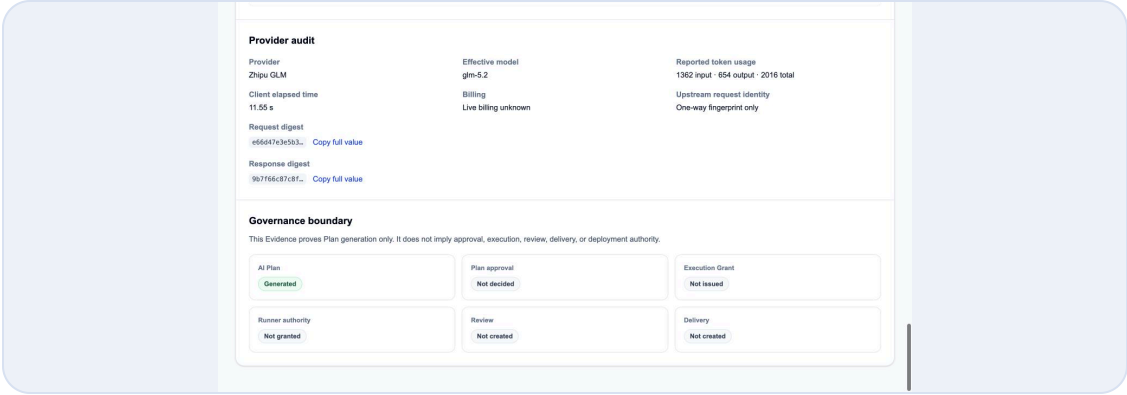
Runner work

Only an eligible, separately granted typed job can reach a Runner-owned fixture workspace.

Grant required

What a Provider cannot do

Issue a grant, run shell commands, mutate a workspace, approve review, authorize delivery, merge or deploy.



PROVIDER LAYER AND BYOK

Support, connection and live validation are separate facts.

Deterministic Fixture is the default

Offline, non-network and non-billable. It is the safest reproducible mode for the product demo.

No credential required | Fixture validated

ONE LIVE-VALIDATED PROVIDER

Zhipu / GLM-5.2

OTHER DISPLAYED ADAPTERS

Supported does not mean live validated

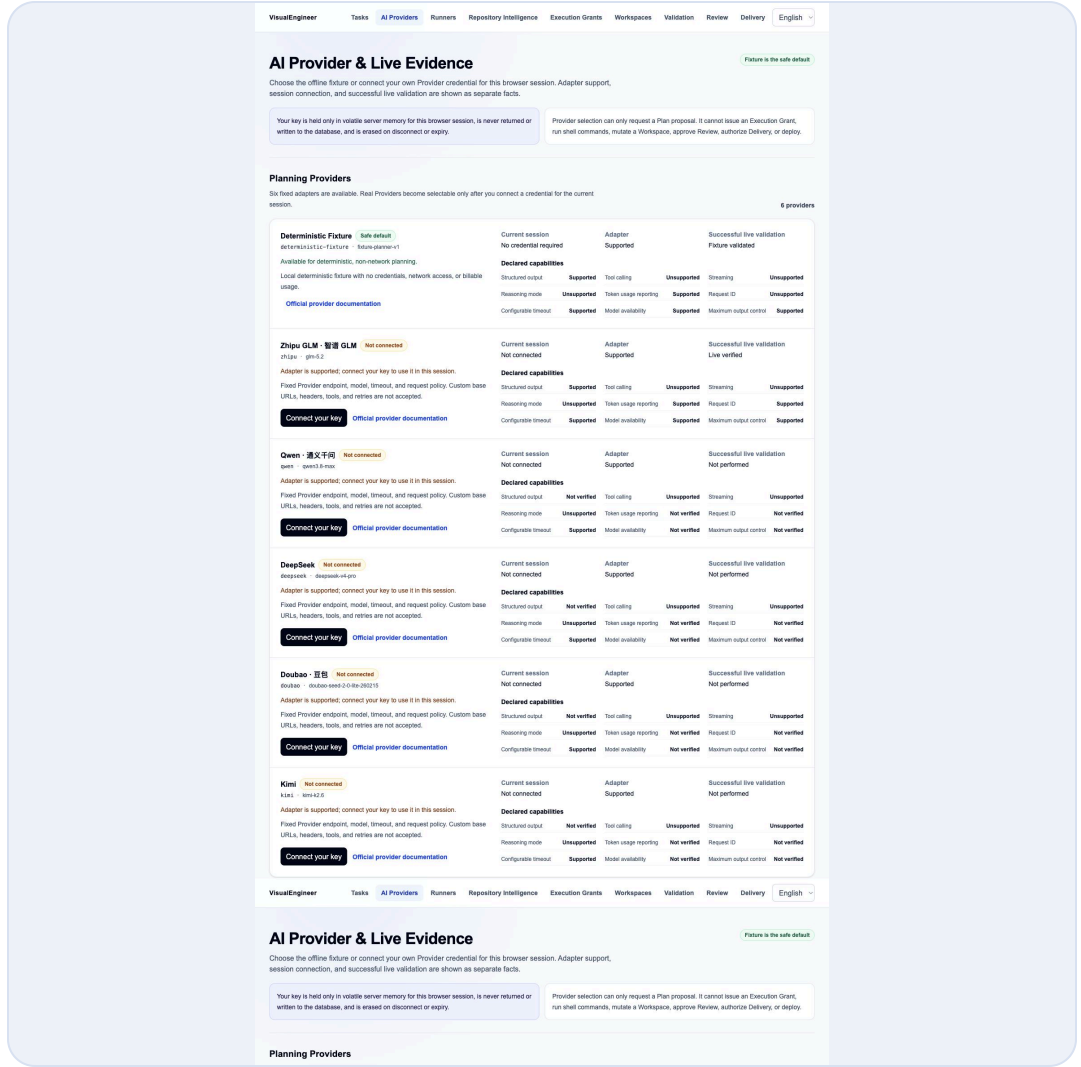
BYOK HANDLING

Current-session volatile server memory

QUOTA BOUNDARY

No ordinary-user fallback to maintainer credentials

> The credential is never returned or persisted and is cleared on disconnect, expiry or restart. This is not represented as a commercial multi-tenant secret vault.



SEALED LIVE VALIDATION

One governed Zhipu / GLM-5.2 invocation succeeded.

CORRELATION

3a9c354f7a58

Safe evidence correlation

REPORTED USAGE

2,016 tokens

1,362 input + 654 output

MEASURED LATENCY

11.55 s

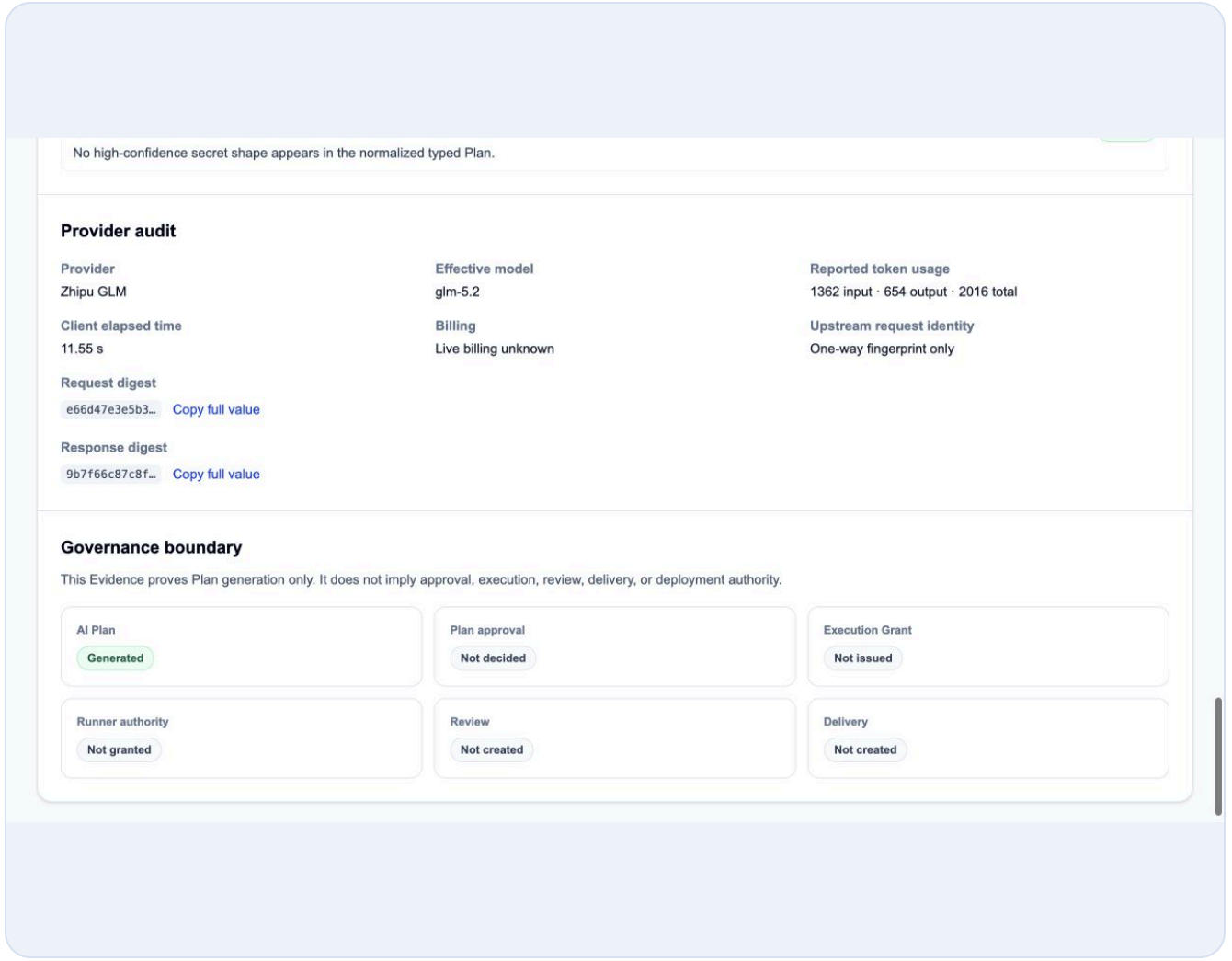
11,543 ms provider; 11,553 ms client total

BILLING

live_billing_unknown

No amount or billing certainty claimed

Lifecycle: succeeded | Typed Plan: 3 operations | 8 policy checks passed



FAILURE ENGINEERING STORY

Unknown after dispatch is a fact, not a failure label to smooth over.

ATTEMPTS 01-02

Transport ambiguity

After dispatch, the system could not honestly determine the Provider outcome.

outcome_unknown

ATTEMPT 03

Response-path diagnosis

A decoding issue was classified truthfully while the earlier immutable history stayed intact.

no auto-retry

NARROW REMEDIATION

Compatibility, not authority expansion

Safe transport observability plus identity content-encoding negotiation and decoded-response normalization.

bounded change

ATTEMPT 04

Separate successful event

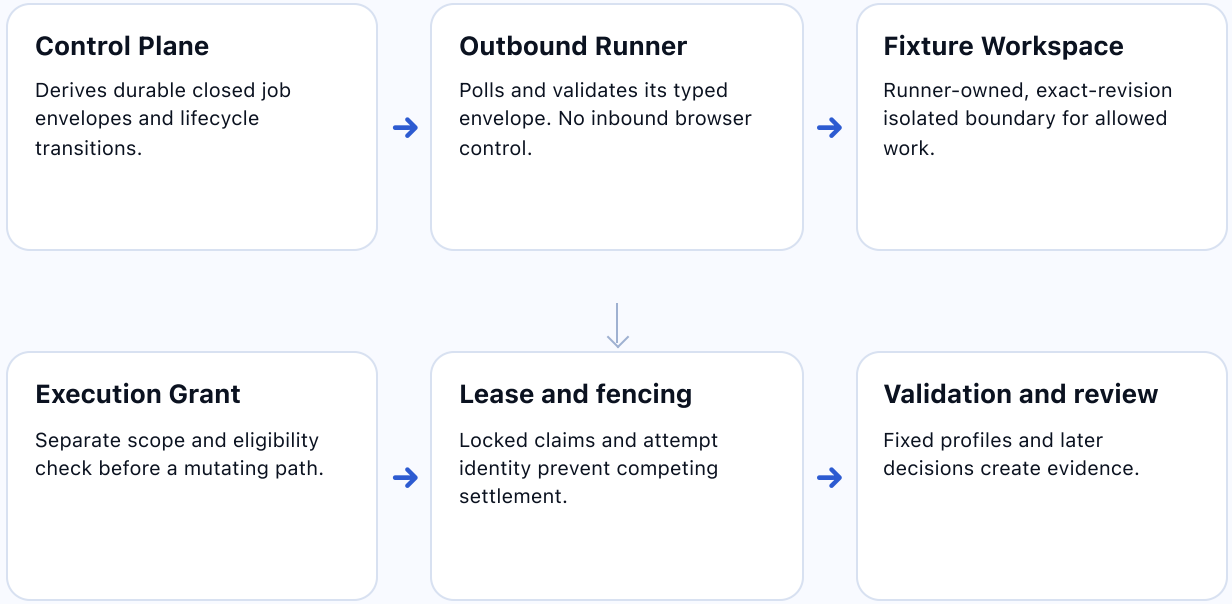
The fourth governed invocation succeeded. It still did not create downstream execution authority.

succeeded

> Why no automatic retry? An ambiguous request may already have reached a billable or side-effecting external system. A later attempt must be a new, controlled event.

RUNNER AND WORKSPACE ISOLATION

The model never receives unrestricted shell authority.



Controls retained at the boundary

- ✓ Closed typed operations rather than arbitrary commands
- ✓ Exact-revision, runner-owned fixture workspace
- ✓ Independent Runner identity and outbound protocol
- ✓ Lease / fencing around job settlement
- ✓ Fixed validation profile and bounded evidence
- ✓ No direct Provider-to-Runner authority path

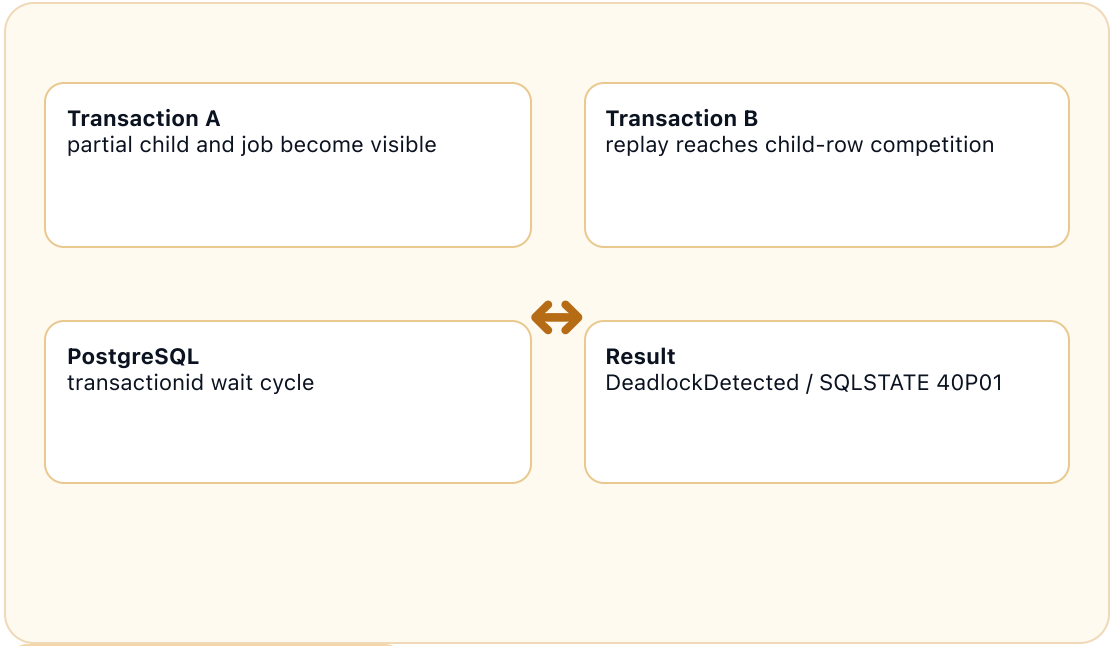
Source repository, remote Git, merge, release and deployment authority are deliberately outside this demo path.

POSTGRESQL CONCURRENCY FAILURE

Fix the atomicity boundary; do not hide a half-write behind retries.

Observed red baseline

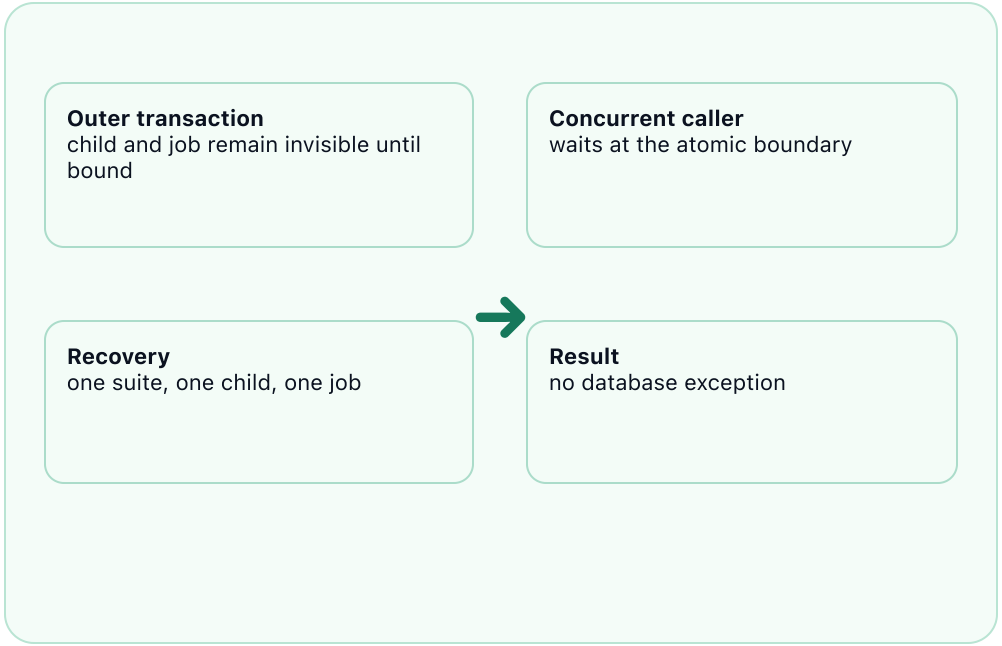
Nested Runner creation committed before the child Validation Run binding was complete. A second transaction could see the partial state and compete for the child row.



10 / 10 deterministic real deadlocks

Controlled remediation

Keep Runner creation inside the outer Validation transaction. A competing transaction waits before it can enter the partially bound-child window.



10 / 10 controlled passes - zero unbound children - no blind retry.

EVIDENCE-FIRST ENGINEERING

Completion claims are backed by a ledger, not a green sentence.

Claims-to-evidence discipline

A claim is accepted only when its source is known and safe to project. Unknowns remain unknown.

Source identity

Git SHA and tree establish the sealed repository state.

Verification

CI, tests, browser evidence and database proof answer different questions.

Provider lifecycle

Prepared, dispatching and terminal classification preserve external uncertainty.

Secret-negative evidence

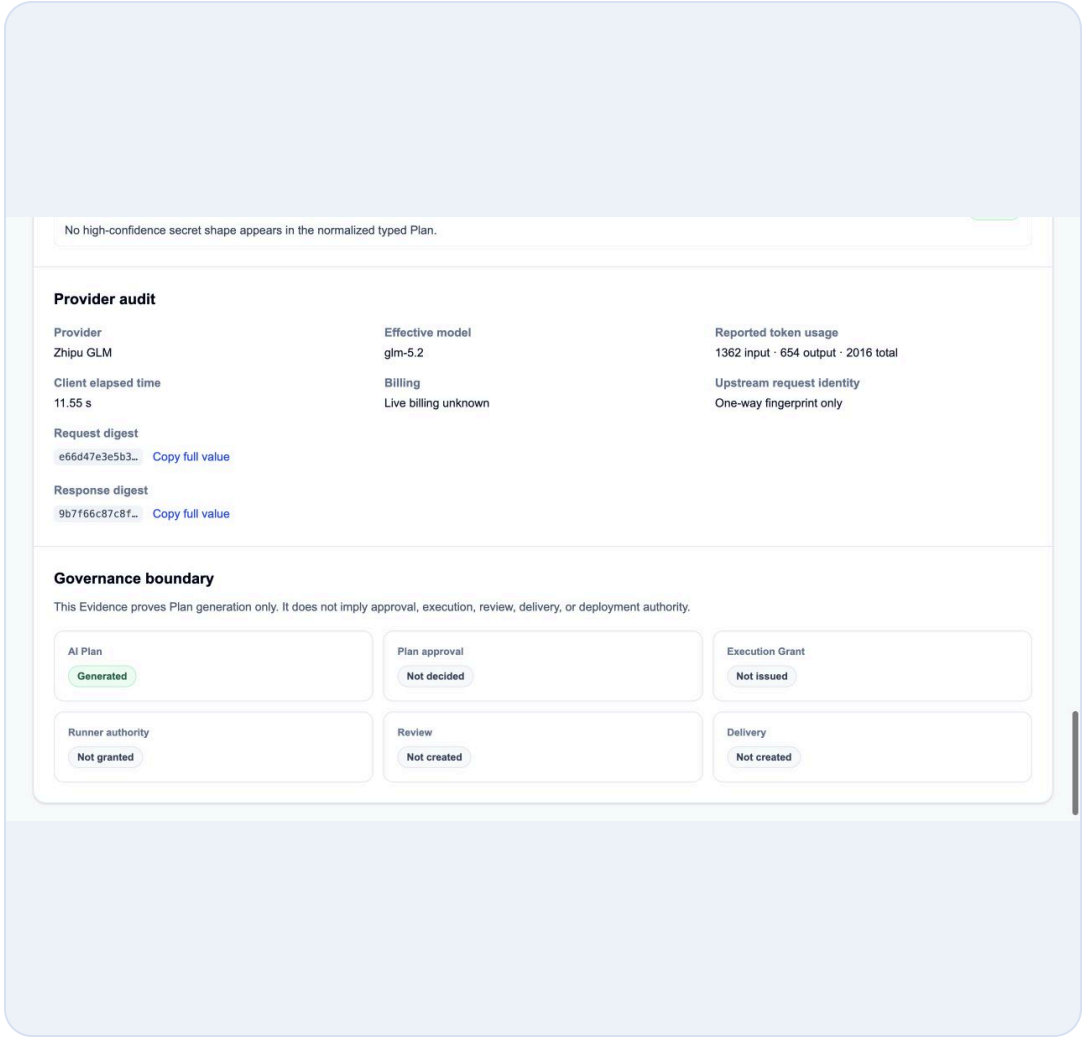
Public-safe material excludes keys, private payloads and account or billing records.

Failure history

Earlier ambiguous attempts remain durable facts, not retroactive success.

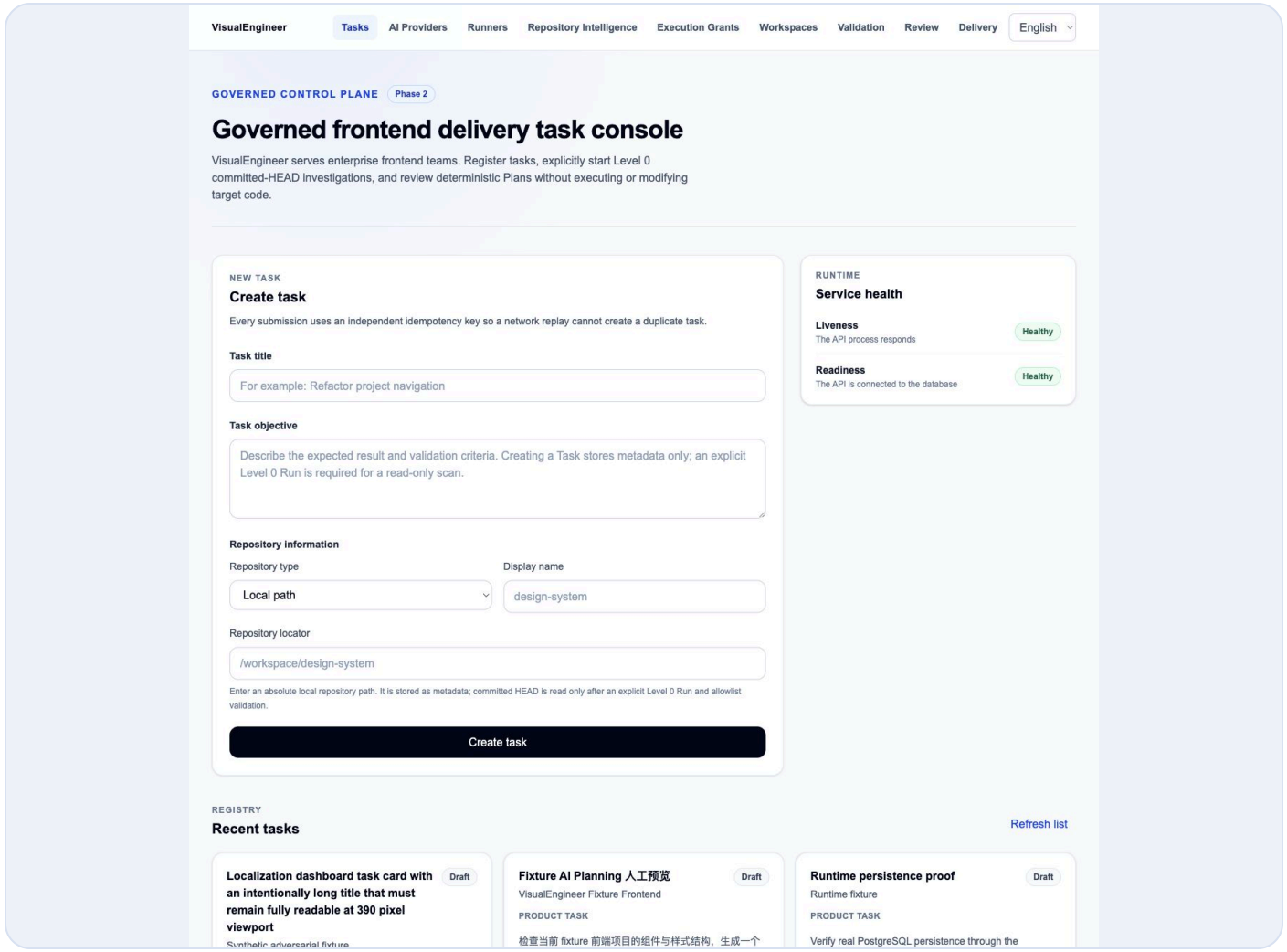
Scope limits

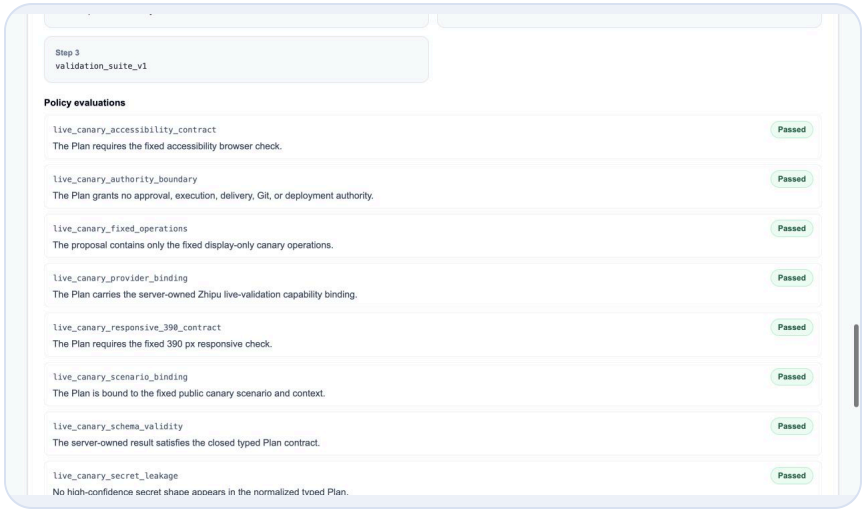
Local demo proof is not restated as production deployment or customer use.

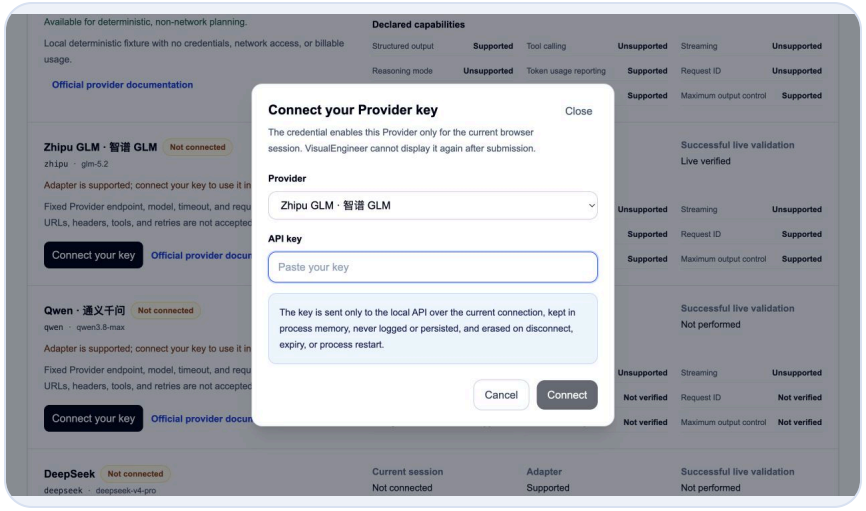


PRODUCT EXPERIENCE

A browser-first console that makes governance visible.







zh-CN / en-US | responsive behavior | typed Plan Evidence | Governance Evidence | Provider status | Fixture / Live distinction | BYOK entry

ENGINEERING STACK

Typed boundaries from browser UI to persistence and proof.

WEB CONSOLE

Next.js App Router
React
TypeScript strict
Tailwind CSS
next-intl

CONTROL PLANE

Python
FastAPI
Pydantic
SQLAlchemy
Alembic

STATE AND EXECUTION

PostgreSQL
Independent Local Runner
Closed typed job protocol
Isolated fixture workspaces

QUALITY AND EVIDENCE

Vitest
Playwright
axe
Ruff and mypy
Git and CI evidence

> The stack matters because every boundary is represented in a concrete contract: TypeScript at the client, OpenAPI at the API edge, typed Python services and PostgreSQL constraints for durable state.

MY ROLE AND AI-ASSISTED ENGINEERING

Accountability for the system boundaries stayed human-owned.

What I owned

PRODUCT	Problem definition, workflow, interface priorities and acceptance criteria.
ARCHITECTURE	Authority boundaries, lifecycle rules, typed contracts and risk decisions.
VALIDATION	Evidence strategy, test expectations, failure interpretation and final acceptance.
EXPLANATION	Ability to discuss the architecture, state machines, failures, tradeoffs and scope limits.

How ChatGPT + Codex accelerated the work

Repository investigation, implementation, tests, documentation and Git / CI work were accelerated with AI assistance under explicit constraints and review.

What this portfolio does not claim

It does not claim that every line was manually written. It does not transfer accountability for product choices, safety boundaries, validation or final acceptance to a model.

> The strongest interview discussion is not "an AI wrote it." It is why the system makes authority, uncertainty and evidence legible.

SCOPE AND NEXT STEP

A clear boundary is a better demo than an overstated claim.

github.com/dankk0220abc-prog/visual-engineer

- Main development is complete; final sealed evidence includes one successful Zhipu / GLM-5.2 live validation.
- Other adapters must not be represented as live validated.
- Fixture remains the safest reproducible demo mode.
- Local, single-user proof only: no production deployment, customer, revenue, user-volume or performance-improvement claim.
- BYOK is session-scoped volatile handling, not a commercial multi-tenant secret vault.

Interview focus: the Provider lifecycle, PostgreSQL invariants, Runner boundary, or evidence strategy.